Publishing 'AI-Driven Design for SIW Filters'

Venue landscape, AI-disclosure requirements, and a recommendation

2026-09-04

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Recommendation for the sole-author paper (updated 2026-09-29)

The draft `ai_driven_rf_design.md` uses both framings. Framing 2 (agent-driven design) is the headline, and framing 1 (the filter, the calibrate-and-invert method and the fabricated board) supplies the evidence. The board is in manufacture, so the timing depends on the measurements.

Step	Venue	Why	When
1	**arXiv** (`eess.SP` primary, `cs.AI` cross-list)	Establishes priority. Closest prior work (arXiv:2608.31006, RF-Agent, RFAmpDesigner) is also on arXiv	As soon as measurements are in; optionally now, as v1
2	**IEEE Access**	Rolling, broad, fast. Accepts a methods paper with hardware and no novel ML model	After measurements
alt.	**Int. J. RF & Microwave CAE** (Wiley)	Exactly on-topic (CAD for RF). Rolling	After measurements
stretch	**IEEE IMS 2027** (4-page)	Flagship, and has an explicit AI/ML-for-RF track. Needs measured data by the deadline	Deadline 6 Jan 2027
watch	**IEEE LAD 2027**	Best topical fit for the agent angle. CFP not yet posted (LAD 2026 closed 2 Mar 2026)	Spring 2027

First choice: arXiv, then IEEE Access, once the measured section is filled in. IMS 2027 is worth attempting only if the boards are measured before mid-December.

AI disclosure is required by every venue above. The draft's Acknowledgment already names the tool, the models and what they did.

What this paper actually is

Two different papers could be written from this project, and they point at different venues:

1. **An SIW filter synthesis paper.** Chebyshev coupling-matrix synthesis, calibrate-and-invert (measure k/Qe on a small closed structure, invert the textbook mapping, assemble and confirm), applied to a dual-mode and an inline-cascade SIW cavity filter, with a fabricated board. This is a normal RF/microwave paper that happens to have been designed with heavy AI assistance.
2. **An agentic-design-automation paper.** An LLM coding agent drove an open-source FEM solver (Palace) through a design loop end to end — wrote the meshing/optimization code, ran the EM solves, generated the fabrication outputs, and in the process produced a documented taxonomy of **fifteen-plus silent modelling bugs**, none of which raised an error. The filter is the case study; the finding is about the process.

These have different homes. Framing (1) fits classic RF/microwave venues. Framing (2) fits the newer, smaller world of venues explicitly about agentic AI and LLM-driven engineering — and that world does **not** currently include the RF/microwave-specific journals, whose "AI for microwave filters" scope almost always means *trained ML surrogate models* (neural networks predicting S-parameters or synthesizing geometry), not an LLM agent writing and iterating the design code itself. Picking the wrong framing for a venue is the single biggest fit risk here.

Do these venues even accept AI-assisted work?

Yes — every venue considered below permits it. None forbid AI assistance; all require **disclosure**, not permission-seeking, and none allow an AI system to be listed as an author.

Publisher	Authorship	Disclosure required	Where
IEEE (LAD, IMS, IMFW, JSTEAP)	AI cannot be an author	Yes, in Acknowledgments: name the tool, name the sections, briefly explain how it was used. Grammar/editing-only use is exempt.	IEEE generative-AI policy, IEEE RAS guidelines
Wiley (IJRFMiCAE)	AI cannot be an author	Yes, full disclosure in a dedicated appendix; author takes full responsibility for the content.	Wiley generative-AI author guidelines
MDPI (Electronics)	AI cannot be an author	Yes, disclosure statement in Acknowledgments naming the tool, plus a Materials-and-Methods description of specifically how it was used. A vague "ChatGPT improved this manuscript" does not meet their standard.	MDPI generative-AI authorship policy

Separately and unconditionally: using AI to fabricate or manipulate data, figures, or results is a research-integrity violation, independent of the disclosure question. Nothing here is at risk on that front — every number in this project traces to a real Palace solve or a real gerber/drill file, and the whole point of the bug taxonomy is that errors were *found and fixed*, not hidden.

Practical takeaway for this specific paper: because the paper's subject is AI-driven design, the disclosure isn't a compliance footnote — it's the paper's own methods section. Be specific about what the agent did (wrote the meshing/optimization/export code, ran the EM solves, found and fixed the bugs described in the results, drafted portions of the text) rather than a blanket statement. That specificity is exactly what MDPI and IEEE both say separates an adequate disclosure from an inadequate one.

Venue options, by framing

Framing 1 — RF/microwave synthesis paper

Venue	Type	Fit	Deadline (from today, 2026-09-04)
Int'l J. of RF and Microwave CAE (Wiley)	Journal, rolling	Exactly on-topic (CAD/EDA for RF/microwave); no deadline pressure	rolling
MDPI Electronics	Journal, rolling, open access	Broad enough, has run ML-for-microwave-filter themes before	rolling
IEEE MTT-S IMS 2027 (San Antonio, May 2027)	Flagship conference	Best prestige match; this cycle explicitly expanded scope for AI/ML-driven RF design	**Jan 6, 2027** — reachable
IEEE JSTEAP "AI for EM/Antennas/Microwave Circuits" special issue	Journal special issue	Ideal scope on paper	this cycle's deadline (Dec 15, 2025) already passed; watch for the next call
IEEE MTT-S IMFW (Filter Workshop, 3-page format)	Workshop	Filter-specific audience, short-paper format suits a methods/negative-result piece	this cycle's (Oct 1, 2025) passed; annual, next around Oct 2026 for IMFW 2027
EuMC / EuMW (London, Oct 2026)	Flagship conference	Good fit	this cycle's deadline (Mar 27, 2026) already passed

Framing 2 — agentic-AI / LLM-driven design-automation paper

Venue	Type	Fit	Deadline (from today)
IEEE LAD ("LLM-Aided Design", `iclad.ai`, IEEE CEDA)	Conference, IEEEXplore	**Best topical match found.** LAD 2026's own stated theme was "agentic optimization and scaling inference-time methods" — a near-exact description of the calibrate-and-invert + local-GP loop this project ran.	LAD 2026 already ran (deadline Mar 2, 2026; conference Jul 30–31, 2026, Stanford). **LAD 2027 not yet posted** — watch `iclad.ai`; historical pattern suggests a spring 2027 deadline.

Venue	Type	Fit	Deadline (from today)
arXiv (`eess.SY` or `cs.AI`)	Preprint, no review	Where the closest comparable paper currently lives (see below)	immediate, no deadline
IEEE Access	Journal, rolling, open access	Broad enough to accept an agentic-AI-methods paper even without a novel trained model; fast review	rolling
IEEE CAI ("Agents for Science and Engineering" session)	Conference special session	Direct topical fit	CAI 2026 already ran (May 2026, Granada); watch for CAI 2027

A directly comparable prior work: *From Prompt to Prototype: Towards a Frontier LLM Driven RF Engineering Workflow* (arXiv:2608.31006, Aug 2026) — an LLM agent operating CST Studio Suite, Keysight ADS, and KiCad via their scripting interfaces to design a GNSS antenna system (patch, SAW prefilter, two-stage LNA) end to end in about two hours. Same category of contribution as this project's, currently only on arXiv with no venue attached — a sign this specific niche (agentic LLM + real EM/CAD tools, not a trained surrogate) does not yet have a single settled home. Worth citing as related work either way.

Recommendation

If the point of the paper is genuinely "an AI coding agent drove an open-source solver through a design loop and the process itself is the finding" (framing 2 — the approach actually followed: Palace, not CST, as the solver of record; AI writing and iterating code, not a trained surrogate model):

1. **Post to arXiv now.** Establishes priority immediately, costs nothing, and is where the one directly comparable paper already lives.
2. **Target IEEE LAD 2027** when its CFP opens (watch `iclad.ai`). This is the venue whose own stated interests (agentic optimization, inference-time scaling, LLM-driven design tools) match the actual contribution most closely of everything found.
3. **IEEE Access** is the fastest realistic peer-reviewed option if IEEE credit is wanted sooner than LAD 2027's cycle allows — it doesn't require a novel ML model, just a real methods contribution, which this has (the bug taxonomy, the calibrate-and-invert method, the fabricated result).

If instead the paper should foreground the filter-synthesis contribution and the fabricated result, with the AI-driven process as a secondary disclosure rather than the headline: **IJRFMiCAE or MDPI Electronics** now (rolling, no deadline pressure), with **IMS 2027** (Jan 6, 2027) as a stretch target for the flagship conference. Nothing prevents submitting both a framing-2 preprint/LAD paper and a framing-1 journal paper from the same underlying work, provided each stands on its own contribution and neither double-publishes the same claims as new.

Sources

- IEEE generative-AI author-disclosure policy
- IEEE RAS guidelines for generative AI usage
- MDPI's generative-AI authorship policy
- Wiley generative-AI guidelines for authors (PDF)
- IEEE LAD 2026 call for papers
- ICLAD / LAD conference series
- From Prompt to Prototype (arXiv:2608.31006)
- IEEE MTT-S IMS 2027 call for papers
- IMFW 2026 call for papers
- EuMW 2026 call for papers
- IEEE JSTEAP AI special issue call
- International Journal of RF and Microwave CAE (Wiley)
- IEEE CAI 2026

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cd docs/paper && pandoc --pdf-engine=xelatex PUBLICATION_VENUES.md -o PUBLICATION_VENUES.pdf
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